

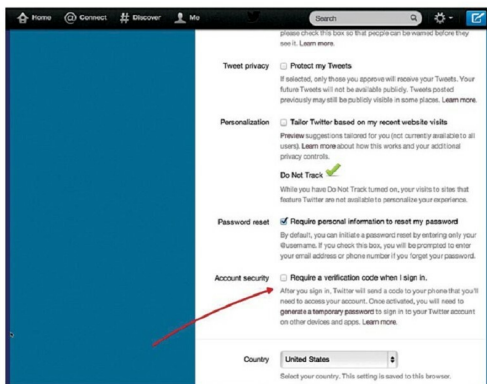


Fig. 12.1: Two-step authentication for a Twitter account (Credit: Twitter)

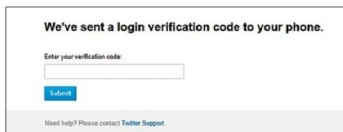


Fig. 12.2: Twitter sends a one-time password to the user's mobile number (Credit: Twitter)



Fig. 13: Two-step authentication for a Facebook account (Credit: Facebook)



Fig. 14.1: Activation of TFA for Microsoft's online services: Outlook, Skype and Xbox (Credit: Microsoft)

cessfully entered the user name and password in the first level. To make sure that it is the same user who entered the first credentials, Twitter sends a six-digit pass code via an SMS to the user's mobile phone. This feature can be activated from the user's Account Settings page, where he or she can check the box for Require A Verification Code When I Sign-Up. The user then needs to enter his or her mobile phone number to use TFA (Figs 12.1 and 12.2).

Facebook

TFA for Facebook can be activated under Settings to protect an account (Fig. 13). If a user has activated TFA successfully, Facebook sends a one-time pass code to his or her mobile phone. By entering the pass code, the user can prove that it is really he or she who is trying to log-in.

Microsoft

Microsoft has already added the option of TFA across its many online services, such as Windows 8/RT, Outlook, Skype, OneDrive, Windows phone and Xbox 360/Xbox LIVE. A user can enable this feature at <https://account.live.com/proofs/Manage>. After enabling this service, Microsoft stops millions of fraud attempts every day.

Fig. 14.1 shows how to activate TFA for Microsoft accounts.

There are two ways in which TFA can rear its head—a security code or an app password. If a user has enabled TFA, after logging-in with Microsoft account credentials (user name and password), you might be prompted to enter a security code or an app password. A security code is needed if a user is using a trusted PC to sign-in to Windows 8 with a Microsoft account. Microsoft will send an SMS that contains the pass code to the user's mobile phone (Fig. 14.2). You must enter this pass code on the website or in Windows.

An app password is needed for those apps or devices that do not work with the security code system.

Google/Gmail

Most users have a lot of data stored in their Google accounts, such as Gmail and Google drive, and it is obvious that they would definitely want to secure it by turning on TFA. Figs 11.1 through 11.3 show how to activate TFA for Google accounts.

Twitter

In May 2013, Twitter introduced TFA to protect user accounts with a more sophisticated log-in system—a new login-verification feature where a user has to enter a six-digit pass code, in addition to the standard password. This pass code (second level) is used after a user has suc-